

WEEKLY ASSIGNMENT • SESSION 2

Build a Cross-Platform Real-Time Application with AI

Apply the concepts from Session 2 by building one simple product across web, Chrome, Android and desktop while exploring real-time communication and HTML Canvas.

Deadline	Sunday, 26 July 2026
Primary Stack	Next.js • Supabase • Cursor AI
Submission	Loom + Vercel URL + Public GitHub

Assignment Goal

The goal is not to build a large product. Build a small, working application that proves you can use AI-assisted development to connect multiple platforms and understand how modern real-time features work.

Session 2 covered: Cursor AI development, authentication, Resend email, 2FA, email templates, Google reCAPTCHA, task CRUD, Supabase database & storage, and a Chrome extension.

Your 7 Tasks

Complete the tasks in sequence. Keep the scope simple, functional and easy to demonstrate.

1

Session 2 - Task 1 - Research Real-Time Communication

Research how real-time communication works in modern web applications.

Understand the basic WebSocket connection lifecycle and use cases.

Understand how Supabase Realtime / real-time database tables work.

Research what changes when a Next.js application is deployed on Vercel and how persistent real-time connections should be handled.

Note: You should be able to explain your chosen real-time approach in the Loom demo.

2

Session 2 - Task 2 - Build a Simple AI-Assisted Web Application

Use Cursor AI to create a small Next.js application backed by Supabase. Choose any useful idea that you can complete within the deadline.

Include authentication and a basic data flow such as add, read, update or delete.

You may reuse concepts from the task management system developed during the session.

Keep the application focused; functionality matters more than visual complexity.

3

Session 2 - Task 3 - Add a Real-Time Feature

Add at least one feature that updates without manually refreshing the page.

Examples: real-time chat, live notifications, online status, live task updates, activity feed or another real-time interaction.

Use Supabase Realtime or another suitable WebSocket-based approach.

Note: A basic working feature is enough. The goal is to understand the concept and integration.

Cross-Platform Build

Use the same core idea across platforms rather than creating four unrelated products.

4

Session 2 - Task 4 - Use HTML Canvas

Add a meaningful feature that uses the HTML Canvas API inside your web application.

Possible ideas include a mini whiteboard, signature pad, drawing tool, chart-like visualization, image annotation, simple game or another Canvas-based interaction.

You may use ChatGPT to brainstorm an idea that naturally combines Canvas with your project.

5

Session 2 - Task 5 - Develop a Chrome Extension

Create a simple Chrome extension related to the same application or project idea.

The extension should perform at least one useful action such as creating/viewing data, quick access, notifications, capturing information or another relevant workflow.

Demonstrate the extension working in Chrome during your Loom video.

6

Session 2 - Task 6 - Create Android and Desktop Versions

Use AI-assisted development to build simple versions of your application for mobile and desktop.

Android: create a minimum working application using React Native with Expo.

Desktop: create a minimum working desktop application using Tauri with Rust.

iOS: optional bonus task for interns who have access to macOS.

Note: Previous native-language expertise is not required, but you should review and understand the AI-generated code and be able to explain the basic flow.

7

Session 2 - Task 7 - Deploy, Record and Propose Your Final Project

Prepare the complete submission and propose the real-world project you want to build during the final two weeks of the internship.

Deploy the web application on Vercel and verify that the live URL works.

Make the GitHub repository public and verify that it opens without requesting access.

Record a Loom video demonstrating the web app, real-time feature, Canvas feature, Chrome extension, Android app and desktop app.

Submit one realistic project idea for the final two weeks. You may discuss it privately with the mentorship team before starting.

Submission Requirements

Your evaluation can only begin when the links are accessible and the functionality is clearly demonstrated.

Required submission items

- Loom video link showing the complete functionality.
- Live Vercel URL for the web application.
- Public GitHub repository link that opens without any access request.
- Final two-week project idea / proposal.

Loom demonstration checklist

- Briefly explain your project idea and architecture.
- Show login/authentication and the main web application flow.
- Show the real-time feature working between two sessions/users where applicable.
- Show the HTML Canvas feature.
- Show the Chrome extension running in the browser.
- Show the Android application running through Expo / emulator / device.
- Show the desktop application running through Tauri.
- Open the public GitHub repository and Vercel deployment in the recording.

Important: Before submitting, test every link in an incognito/private window. Restricted GitHub repositories, Drive files or inaccessible Loom/Vercel links delay evaluation and may affect timely review.

Deadline

Sunday, 26 July 2026

Keep the project simple, working and understandable. Use AI as your development assistant, but make sure you understand what you build and can explain it during evaluation.